Qihao Qian

q2qian@ucsd.edu | Personal Website

Education

University of California, San Diego (UCSD) **Sep 2024 – Dec 2026**
Master of Science in Electrical and Computer Engineering *San Diego, US*

University of Electronic Science and Technology of China (UESTC) **Sep 2020 – June 2024**
Bachelor of Engineering in Software Engineering *Chengdu, China*

Industry Experience

Tiktok (ByteDance) **May 2026 – Aug 2026**
MLE Intern *Beijing, China*

- Led post-training of **Qwen3-1.7B/4B/8B** for TikTok Tako query suggestion generation using training data constructed from millions of users' long- and short-term memories. Built the end-to-end training pipeline with **LLaMA-Factory**, **DeepSpeed ZeRO-3**, and **vLLM** (8×H100), resulting in **two production launches**.
- Built a preference optimization pipeline with **DPO**, reward-model filtering, trending-aware supervision, and diversity-aware data selection, improving **Inbox Query Recommendation PV CTR by **% (US)** and ****% (ROW)**.
- Designed an offline evaluation framework covering structured-output validity, semantic diversity, and standard LLM benchmarks (**MMLU**, **HellaSwag**, **IFEval**); demonstrated that **single-stage ORPO** outperformed the SFT→DPO pipeline while reducing **training compute by 50%**, and further explored **On-Policy Distillation (OPD)**.
- Reduced feature tokens by **51.5%** through semantic feature injection, improved small-model robustness via training and data-quality optimization, and enabled a production-ready **4B real-time serving model**.

Xbrain(Early Stage Startup) **Feb. 2024 – June 2024**
MLE Intern *Nanjing, China*

- Built an audio-driven digital-human lip-sync system using U-Net + PatchGAN, trained with multi-GPU DDP.
- Built a **Ray**-based distributed data preprocessing pipeline covering face detection (**RetinaNet**), speech enhancement (**FRCRN/Demucs**), mel-spectrogram extraction, and **SyncNet** audio–video temporal alignment filtering, enabling automated cleaning and filtering of **1000+ hours** of training data.
- Designed a composite loss combining **L1** reconstruction, **VGG** perceptual loss (**LPIPS**), **PatchGAN** adversarial loss, and **SyncNet** lip-sync loss, with three-stage progressive training (L1+LPIPS → +GAN → +SyncNet).
- Compared to Baseline-Wav2Lip, achieved **10.2%** lower LSE-D, **20.5%** higher LSE-C, and **12.1%** lower FID.

DJI **June 2022 – Dec 2022**
MLE Intern *Shenzhen, China*

- Improve the **30-minute per scene** bottleneck in autonomous-driving lane annotation to **under 5 minutes** by building an AI-assisted platform where users annotate lanes via rough bounding boxes.
- Trained **PointNet++** on Toronto3D, and introduced **Random Drop**, **Global Scaling & Rotation**, and lane **Oversampling** data augmentation to handle sparse real-world lane point clouds.
- Achieved lane segmentation **F1 85.3%** on the test set; built a GPU inference service based on **PyTorch** with custom **CUDA extension** and deployed with **Docker**, reliably supporting large-scale internal batch pre-annotation tasks.

Research Experience

Online Euclidean SDF Mapping **Jan 2025 – Mar 2026**
UCSD Existential Robotics Lab | ***IROS 2026*** *(First Author)* | ***AutonomousRobots*** *(Third Author)* *San Diego, US*

- Targeting the challenges in Online ESDF Mapping—balancing real-time performance, memory efficiency, and geometric accuracy while providing effective supervision in free space—led the proposal and implementation of the OREN (Prior + Residual) framework, enabling efficient, accurate, and spatially continuous ESDF mapping.
- Designed a **semi-sparse octree** that improves memory efficiency and interpolation accuracy (**-45%** SDF MAE, **-38%** Grad. MAE) compared with the sparse octree, enabling faster training and smoother surface reconstruction.
- Proposed a **projection loss** that provides supervision for free-space regions, improving reconstruction accuracy by over **6×** in Chamfer distance and **+14%** in F1 score on the Replica dataset while achieving faster convergence.
- Implemented all the training and evaluation pipeline and conducted extensive ablation and baseline experiments, demonstrates that **OREN** outperforms the state of the art in terms of accuracy and efficiency.
- Validated the model on the challenging Newer College dataset and successfully **deployed the pipeline on a Clearpath Jackal robot** and in simulation, demonstrating robust real-world performance in complex environments.